

Rubber-Band Dynamics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Valence Gap and Re-Alignment Analysis

PENPAL Project

2026-05-01

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1 Overview

This notebook compares **rubber-band dynamics** across conditions using the exchange-aligned data structure.

The main quantity is the **valence gap**:

- `valence_gap = slot_1_valence - slot_2_valence`
- `gap_change = current_gap - previous_gap`

A negative slope in `gap_change ~ previous_gap` indicates a rubber-band effect: when the slots diverge, the next aligned exchange tends to pull them back together.

As in the updated valence notebook:

- `author_1 / author_2` are treated as **slot 1 / slot 2**
- the analysis is restricted to **complete exchanges**

- the temporal window uses `analysis_turn / turn_index`, not the old raw-turn logic

Primary cross-condition object. For each story i , fit the OLS

$$\text{gap_change}_{i,t} = \alpha_i + \beta_i \cdot \text{valence_gap_lag1}_{i,t} + \varepsilon_{i,t}$$

and define the per-story reversion strength as $R_i = -\beta_i$, so that larger positive R_i means a stronger pull back together. R_i is already label-invariant (the gap enters squared/reversal-style in the relationship). We test $R_i \sim \text{condition}$ with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. The pooled LMM slopes are kept as supplemental.

Article-use note. This notebook tests a dyad-level regulation mechanism, not an agent-level asymmetry. If R_i does not differ by condition, the conclusion is that emotional re-convergence is a general co-writing dynamic rather than a Human-AI-specific asymmetry.

2 Load Data

Table 1: Loaded valence data for rubber-band analysis

condition	n_stories	n_turns
Human-AI	100	751
Human-Human	36	288
AI-AI	40	320

3 Preprocessing

Analysis rows: 1183

Stories: 176

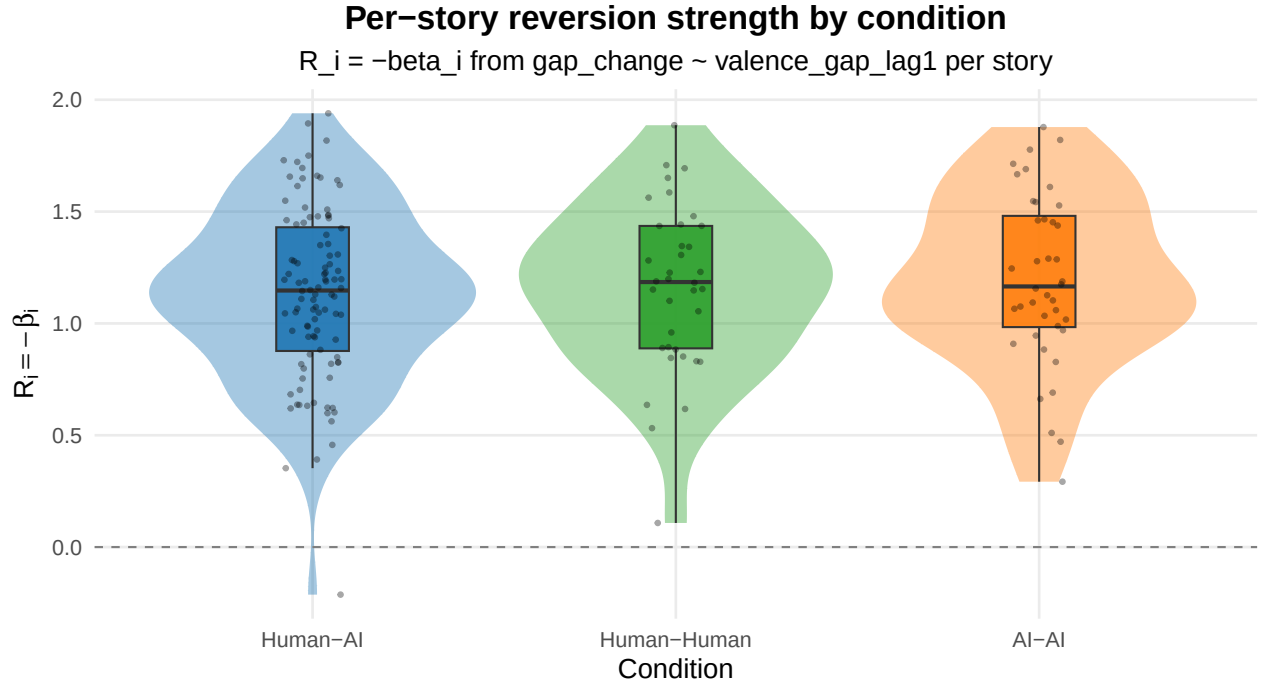
4 Primary: Per-Story Reversion Strength

Table 2: Per-story reversion strength $R_i = -\beta_i$ by condition

condition	n	mean_R	median_R	sd_R
Human-AI	100	1.132	1.147	0.383
Human-Human	36	1.157	1.185	0.374
AI-AI	40	1.198	1.165	0.382

```
## One-way ANOVA: R ~ condition
##           Df Sum Sq Mean Sq F value Pr(>F)
## condition    2  0.126  0.06324   0.435  0.648
## Residuals  173 25.139  0.14531
##
## Kruskal-Wallis (robust): R ~ condition
##
## Kruskal-Wallis rank sum test
##
## data: R by condition
## Kruskal-Wallis chi-squared = 0.8088, df = 2, p-value = 0.6674
##
```

```
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: per_story_rb$R and per_story_rb$condition
##
##           Human-AI Human-Human
## Human-Human 0.7      -
## AI-AI        0.7      0.7
##
## P value adjustment method: BH
```



5 Bootstrap CIs and Effect Sizes on R_i

Table 3: Bootstrap 95% CI on per-condition mean of reversion strength R_i ($R = 10,000$)

condition	n	mean	ci_lower	ci_upper
Human-AI	100	1.132	1.057	1.205
Human-Human	36	1.157	1.033	1.275
AI-AI	40	1.198	1.079	1.313

Table 4: Pairwise Cliff's delta on R_i

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	100	36	-0.049	negligible
Human-AI vs AI-AI	100	40	-0.094	negligible

contrast	n_a	n_b	cliff_delta	magnitude
Human-Human vs AI-AI	36	40	-0.053	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 5: Planned contrast: R_i in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	-0.046	0.058	-0.79	160.264	0.43	-0.16	0.069

7 Time-Resolved Reversion Strength

Table 6: Per-story reversion strength by condition and story phase

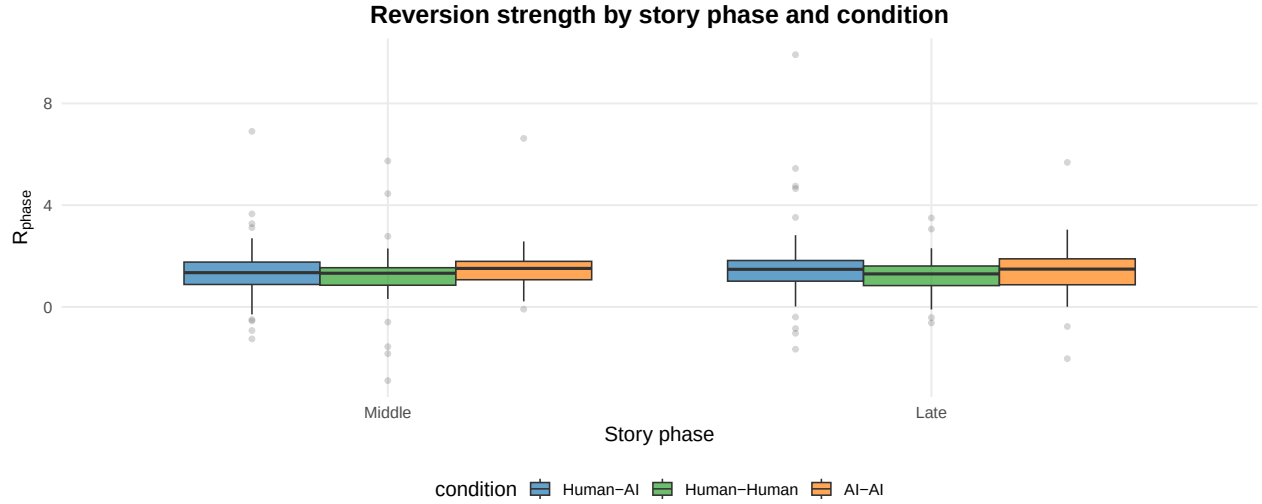
condition	phase	n	mean_R	sd_R
Human-AI	Middle	98	1.322	1.005
Human-AI	Late	99	1.520	1.314
Human-Human	Middle	36	1.169	1.473
Human-Human	Late	36	1.257	0.882
AI-AI	Middle	40	1.566	1.013
AI-AI	Late	40	1.401	1.189

```
## LMM: R_phase ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: R_phase ~ condition * phase + (1 | conversation_id)
## Data: per_story_phase_rb
##
## REML criterion at convergence: 1100.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4402 -0.4097 -0.0022  0.3031  6.9436
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.1006   0.3172
## Residual                1.2563   1.1208
## Number of obs: 349, groups: conversation_id, 176
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)      1.3214     0.1177 341.2938  11.230 <2e-16
## conditionHuman-Human -0.1524     0.2270 341.1884  -0.671  0.502
## conditionAI-AI       0.2450     0.2186 341.1915   1.121  0.263
## phaseLate          0.1992     0.1598 173.6362   1.247  0.214
## conditionHuman-Human:phaseLate -0.1114     0.3088 171.7159  -0.361  0.719
```

```

## conditionAI-AI:phaseLate      -0.3651      0.2972 171.7711  -1.228      0.221
##
## (Intercept)                  ***
## conditionHuman-Human
## conditionAI-AI
## phaseLate
## conditionHuman-Human:phaseLate
## conditionAI-AI:phaseLate
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H cndAI-AI phasLt cH-H:L
## cndtnHmn-Hm -0.518
## condtnAI-AI -0.538  0.279
## phaseLate   -0.683  0.354  0.368
## cndtnHm-H:L  0.353 -0.681 -0.190  -0.518
## cndtAI-AI:L  0.367 -0.190 -0.681  -0.538  0.278

```

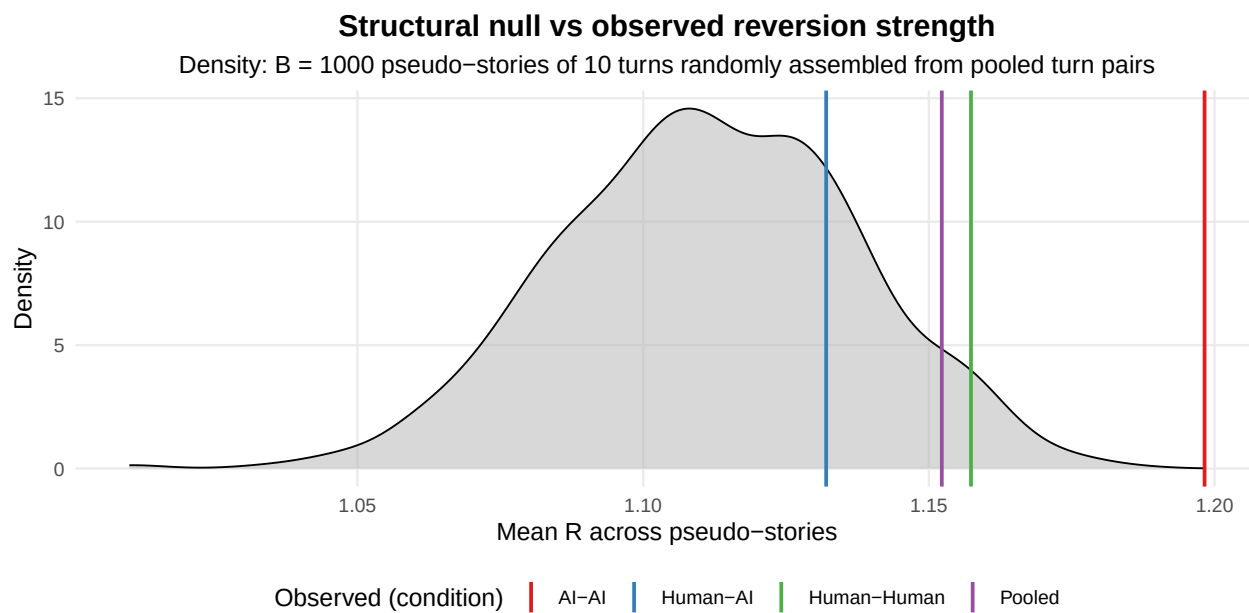


8 Structural-Null Permutation Test

Motivation: the observed rubber-band slope is near-identical across HH, HA, and AA, suggesting a structural / mean-reversion artifact of the `gap_change ~ gap_lag1` parameterization rather than genuine coordination. This test pools all turn-level (slot1, slot2) valence pairs across all stories and conditions, randomly repartitions them into pseudo-stories of 10 turns, and recomputes per-pseudo-story $R = -\beta$. If the observed mean R falls inside the resulting null distribution, the rubber-band effect is structural; if observed exceeds the null, it reflects genuine dyadic coordination.

Table 7: Structural-null test: observed mean R per condition vs 1000-iter null (pseudo-stories of 10 turns from 1359 pooled pairs)

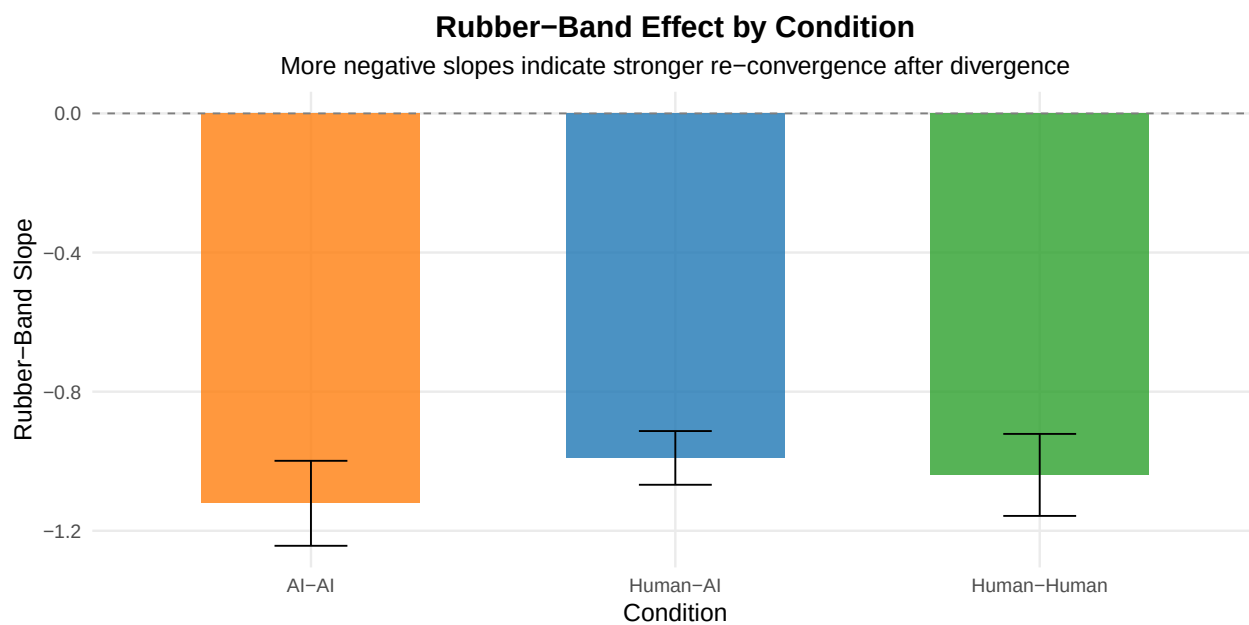
condition	observed_mean_R	null_mean	null_sd	p_perm_two_sided
Human-AI	1.1320	1.1115	0.0261	0.4426
Human-Human	1.1574	1.1115	0.0261	0.0779
AI-AI	1.1982	1.1115	0.0261	0.0030
Pooled	1.1523	1.1115	0.0261	0.1219



9 Supplemental: Pooled Condition-Specific Slopes

Table 8: Rubber-band slopes by condition

condition	estimate	se	t	p	sig
Human-Human	-1.039	0.060	-17.280	0	***
Human-AI	-0.990	0.039	-25.139	0	***
AI-AI	-1.121	0.062	-17.986	0	***



9.1 Supplemental: Cross-Condition Interaction Model

Cross-condition model summary:

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: gap_change ~ valence_gap_lag1 * condition + (1 | conversation_id)
## Data: df_analysis
##
## REML criterion at convergence: 1246.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2694 -0.6064 -0.0215  0.6059  3.0076
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.01068  0.1034
## Residual                0.15570  0.3946
## Number of obs: 1183, groups: conversation_id, 176
##
## Fixed effects:
##                                     Estimate Std. Error      df t value
## (Intercept)                      -0.10259    0.01904  139.94418  -5.389
## valence_gap_lag1                  -0.96537    0.03790 1160.77924 -25.475
## conditionHuman-Human               0.10793    0.03574  123.10875   3.020
## conditionAI-AI                    0.11871    0.03445  123.82111   3.445
## valence_gap_lag1:conditionHuman-Human -0.08351    0.06875 1166.16434  -1.215
## valence_gap_lag1:conditionAI-AI     -0.20088    0.08214 1168.86903  -2.446
##                                     Pr(>|t|)
## (Intercept)                      2.93e-07 ***
## valence_gap_lag1                  < 2e-16 ***
## conditionHuman-Human              0.003074 **
## conditionAI-AI                    0.000779 ***
## valence_gap_lag1:conditionHuman-Human 0.224750
## valence_gap_lag1:conditionAI-AI     0.014610 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) vln__1 cndH-H cAI-AI v__1:H
## vlnc_gp_lg1  0.210
## cndtnHmn-Hm -0.533 -0.112
## condtnAI-AI -0.553 -0.116  0.294
## vlnc__1:H-H -0.116 -0.551  0.051  0.064
## vl__1:AI-AI -0.097 -0.461  0.052  0.023  0.254
##
## Condition-specific slopes:
## condition valence_gap_lag1.trend      SE    df lower.CL upper.CL
## Human-AI          -0.965 0.0380 1166      -1.04   -0.891
## Human-Human        -1.049 0.0574 1148      -1.16   -0.936
## AI-AI              -1.166 0.0730 1162      -1.31   -1.023
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##

```

Pairwise slope contrasts:

```
## contrast estimate SE df t.ratio p.value
## (Human-AI) - (Human-Human) 0.0835 0.0688 1170 1.213 0.6762
## (Human-AI) - (AI-AI) 0.2009 0.0823 1171 2.442 0.0443
## (Human-Human) - (AI-AI) 0.1174 0.0929 1157 1.264 0.6195
```

##

Degrees-of-freedom method: kenward-roger

P value adjustment: bonferroni method for 3 tests

10 Supplemental: General Gap Structure

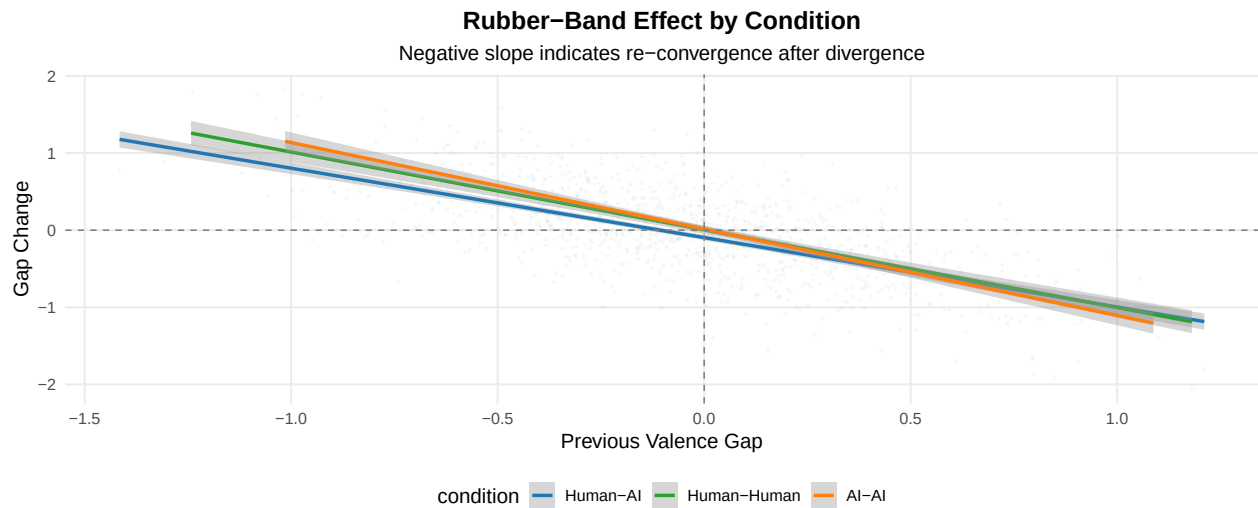
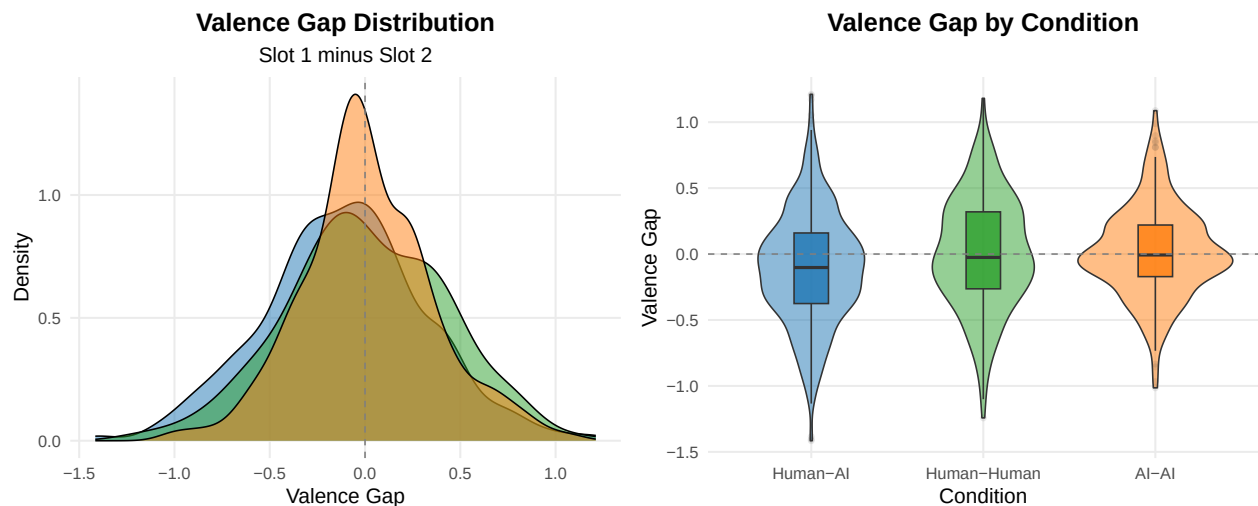
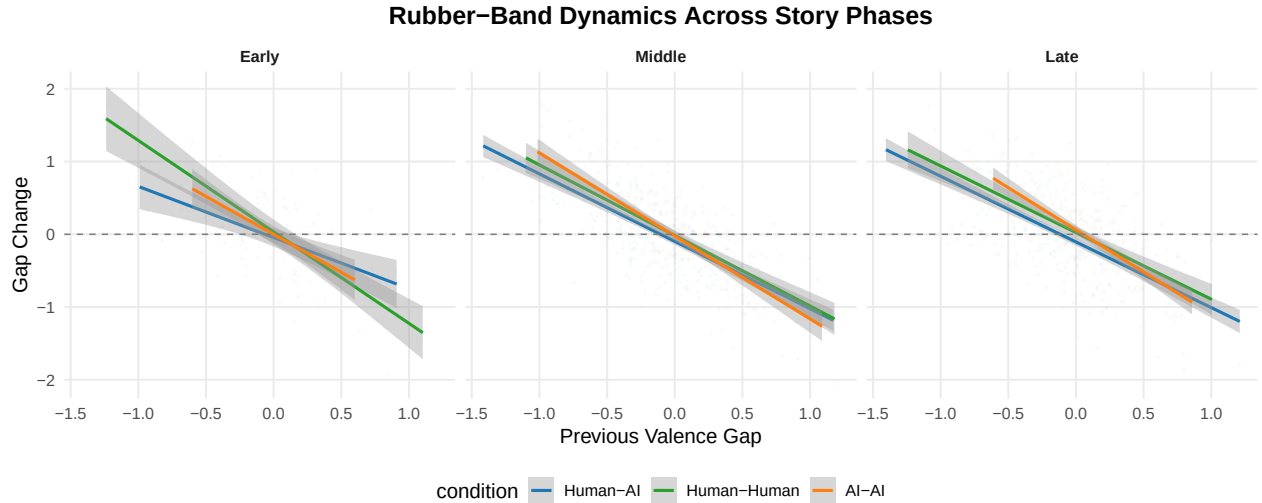


Table 9: Gap and gap-change descriptives by condition

condition	n_turns	mean_gap	sd_gap	mean_gap_change	sd_gap_change	rubber_band_r
Human-AI	651	-0.107	0.426	-0.001	0.570	-0.670
Human-Human	252	0.005	0.424	-0.003	0.616	-0.725
AI-AI	280	0.013	0.347	-0.003	0.508	-0.733





11 Summary

Primary cross-condition result: per-story reversion strength $R_i = -\beta_i$ from `gap_change ~ valence_gap_lag1`, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of R_i
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- time-resolved view of R across early / middle / late story phases
- **structural-null permutation test**: shuffles all turn-level valence pairs across stories and conditions, repartitions into pseudo-stories, and compares observed mean R to the resulting null. Tests whether the rubber-band effect is a coordination dynamic or a mathematical / mean-reversion artifact

Supplemental views retained:

- pooled within-condition LMM slopes (population-level rubber-band estimate per condition)
- cross-condition interaction LMM (`gap_change ~ valence_gap_lag1 * condition + (1|story)`)
- gap distributions and story-phase splits

The Human-AI rubber-band decomposition lives in `analysis/human-ai/rubber_band_analysis.Rmd` and is not duplicated here.

Deferred to a follow-up pass:

- starter-mechanism analysis of the reversion object

Analysis completed: 2026-05-01 23:16:15.778947